Supplementary Material:
Regime-dependent advantage in quantum physics-informed neural networks: a pre-registered comparison on an ODE/PDE hardness ladder

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CONTENTS

I. CIRCUIT SPECIFICATIONS AND FAITHFULNESS MANIFEST

The four quantum-PINN ansatz families used as solver/forecaster encoders in the main text are each implemented against published specifications; the binding manifest is `refs/CIRCUIT_SPECS.md` in the open-source release. Table I summarizes the per-family configuration used throughout the paper.

A. Cell-level dynamics, all variants

The hidden-state dynamics inside the Diffrax integration step (for the forecaster task; the solver task does not integrate a hidden state) are:

$$\text{LiquidQuantumCell:} \quad \frac{dh}{dt} = -\left(\frac{1}{\tau} + q(x)\right) \odot h + A \odot q(x) \tag{1}$$

$$\text{NonLiquidQuantumCell:} \quad \frac{dh}{dt} = -q(x) \odot h + A \odot q(x) \tag{2}$$

$$\text{ClassicalLTCCell:} \quad \frac{dh}{dt} = -\frac{1}{\tau} \odot h + \text{MLP}([h, x]) \tag{3}$$

$$\text{PlainNeuralODECell:} \quad \frac{dh}{dt} = \text{MLP}([h, x]) \tag{4}$$

The two pairs (LQC vs NLQC) and (LTC vs PlainNeuralODE) differ by exactly the $-(1/\tau) \odot h$ term. The cell-level math-identity test (`tests/qlnn_/test_non_liquid_quantum_cell.py`) verifies this identity numerically at random probe points.

II. PER-CELL ERROR TABLES, FULL PRECISION

A. Solver-task ODE matrix ($4 \times 4 \times 3 = 48$ cells)

Table II reports relative-L^2 error per (family, system, seed) at full 4-decimal precision. Best-per-row quantum family is bolded.

B. Solver-task PDE matrix ($4 \times 4 \times 3 = 48$ cells)

C. Multi-family solver matrix

The §2.1 and §2.2 tables aggregate per-cell relative-L^2 numerically across the ODE and PDE solver matrices. For readers who want the visual cross-check, this subsection adds two figures: the peer-review audit iteration that drove the audit-corrected configurations, and the multi-family quantum-PINN matrix visualization spanning four SOTA ansatz families on three PDEs from the hardness ladder. Both figures source data directly from the JSONs underpinning the tables above.

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TABLE I. Quantum-PINN ansatz roster. Per-scalar-circuit parameter counts. Ansätze marked “2D port” add a split-qubit coordinate encoding to the same base circuit (P3.9 design).

Family	Source	Cell paradigm	PQC params	Classical params	Notes
`data_reuploading`	Pérez-Salinas 2020	encoder + LiquidQuantumCell	60	0	4 qubits, 5 layers (ODE
`hardware_efficient`	registry default	encoder + LiquidQuantumCell	60	0	ring entanglement
`strongly_entangling`	PennyLane native	encoder + LiquidQuantumCell	60	0	alias of data_reuploadin
`brickwall`	registry default	encoder + LiquidQuantumCell	60	0	adjacent-pair CZ patter
`te_qpinn_fnn`	Berger 2025	FNN encoder + 4-qubit PQC	60	100	2D port: `te_qpinn_fnn_`
`te_qpinn_qnn`	Berger 2025	QNN encoder + 4-qubit PQC	84	0	zero classical params; 2
`qcpinn`	Farea 2025	Classical pre/post + 4-qubit PQC	15	706	largest classical overhea
`chebyshev_dqc_2d`	P3.7 design	2D Chebyshev tower + HEA	–	0	solver-task only, $n_t = n$
`rf_qrc`	Ahmed 2024	Frozen reservoir + ridge readout	0 (trainable)	varies	non-liquid by constructi
`non_liquid_<ansatz>`	this work (P7.11)	encoder + NonLiquidQuantumCell	same as base $-Q$	same	τ-ablated; 4 variants

TABLE II. Solver-task ODE per-cell relative-L^2 error (PDEBench / FNO convention). Each row corresponds to one (system, ansatz, qubits, depth) configuration evaluated at a single random seed; each column reports one of the four quantum families plus the classical PINN baseline at the matched training budget. Bold marks the per-row best quantum entry.

System	Seed	cheb	fnn	qnn	qcp	cPINN
Lotka–Volterra	0	0.0996	0.0879	0.5242	**0.0074**	0.0117
Lotka–Volterra	1	0.1161	0.1407	0.5229	**0.0029**	0.0078
Lotka–Volterra	2	0.1024	0.1409	0.5240	**0.0070**	0.0071
Van der Pol	0	0.9064	**0.8213**	1.1796	3.4025	1.1569
Van der Pol	1	1.1516	0.8652	**0.7622**	0.8674	0.9434
Van der Pol	2	0.9087	**0.8176**	1.1890	2.6739	1.0550
Lorenz	0	1.0013	0.9950	**0.9794**	0.9957	0.9958
Lorenz	1	0.9994	0.9947	**0.9769**	0.9957	0.9957
Lorenz	2	0.9966	0.9946	**0.9773**	0.9943	0.9957
FitzHugh–Nagumo	0	0.5311	0.4521	**0.3270**	1.2126	0.6398
FitzHugh–Nagumo	1	0.6152	0.3903	**0.3560**	1.5467	0.6410
FitzHugh–Nagumo	2	0.6165	0.6248	**0.3290**	1.6251	1.3265

D. Forecaster-task complete 2×2 matrix ($n = 9$)

The body’s Table **??** reports the all-to-all 5-quantum-family × 4-classical-baseline matrix. Here we add the four non-liquid quantum forecaster columns (P7.11) so the full 2×2 decomposition is visible at the per-cell level.

III. BOOTSTRAP DIAGNOSTICS AND GUARD REPORTS

All paired-bootstrap intervals are computed via `src/quantum_liquid_neuralode/evaluation/h1_verdict.py` with $n_{\text{iter}} = 10{,}000$, $\alpha = 0.05$ (95% percentile CI), and a fixed seed for reproducibility. The skyline guard uses threshold 0.5 (amendment A1) and excludes cells where the known-RHS least-squares fit cannot reach the threshold.

A. Per-bin sample counts (post-guard)

Table V reports the kept-sample sizes per regime bin for each headline verdict.

B. Algebraic identities, exact

Two per-cell identities, verified by the integrity script with tolerance 10^{-3}:

$$\begin{aligned}\Delta_{\text{combined}} &= \Delta_{\text{quantum\_via\_LTC}} + \Delta_{\text{liquid\_via\_classical}} \\ &= -0.6160 + 0.1153 = -0.5007\checkmark \end{aligned} \tag{5}$$

$$\begin{aligned}\Delta_{\text{combined}} &= \Delta_{\text{liquid\_via\_quantum}} + \Delta_{\text{quantum\_via\_nonliquid}} \\ &= -0.3339 + (-0.1668) = -0.5007\checkmark \end{aligned} \tag{6}$$

Both identities are derived from the same per-cell records; only the slot-assignment in the bootstrap differs. The exactness ($|\text{diff}| < 10^{-3}$) is the cleanest verification that the decomposition is a real mathematical structure, not a sampling artifact.

C. Q-Q residual diagnostics (post-hoc; see Amendment A14)

The diagnostic figure suite includes a residual-distribution panel (`fig_residual_analysis`, middle column) whose original source comment mis-labelled it as a Q-Q plot. The implementation is actually a histogram

TABLE III. Solver-task PDE per-cell relative-L^2 error (PDEBench / FNO convention). Each row corresponds to one (system, ansatz, qubits, depth) configuration evaluated at a single random seed; each column reports one of the four 2D-port quantum families plus the classical PINN baseline at the matched training budget. Bold marks the per-row best quantum entry; "—" marks runs that diverged with no finite L^2 measurement.

PDE	Seed	cheb_2d	fnn_2d	qnn_2d	qcp_2d	cPINN
Heat	0	0.0553	0.0023	0.0455	**0.0010**	0.0054
Heat	1	0.0560	0.0031	0.0455	**0.0020**	0.0046
Heat	2	0.0565	**0.0013**	0.0458	0.0020	0.0036
Burgers smooth	0	0.3475	0.0146	0.3647	**0.0112**	0.0428
Burgers smooth	1	0.3522	**0.0126**	0.3543	0.0160	0.0191
Burgers smooth	2	0.3742	0.0246	0.3556	**0.0206**	0.0187
Allen–Cahn	0	0.5783	0.5514	**0.0528**	0.8956	0.2018
Allen–Cahn	1	0.5200	0.5345	**0.0489**	0.5620	0.0633
Allen–Cahn	2	0.6146	0.0927	**0.0534**	0.4936	0.0514
Burgers shock	0	0.4282	0.2507	0.4268	**0.2408**	0.1682
Burgers shock	1	0.4385	0.1589	0.4227	**0.1526**	0.2730
Burgers shock	2	0.4168	**0.1630**	0.4473	0.2553	0.1270

TABLE IV. Forecaster-task non-liquid quantum cells (P7.11). Same training budget and adapter as the liquid forecasters in the body's Table **??**; only the cell's τ-leak removed.

System	Seed	nl_dataru	nl_hweff	nl_stngent	nl_brick
Lotka–Volterra	0	0.5830	0.6455	0.5830	0.7732
Lotka–Volterra	1	0.7951	0.7196	0.7951	0.7057
Lotka–Volterra	2	0.6482	0.6791	0.6482	0.3088
Van der Pol	0	1.4067	1.2307	1.4067	—
Van der Pol	1	9.6720	—	9.6720	—
Van der Pol	2	1.2035	—	1.2035	—
Lorenz	0	1.2807	1.2699	1.2807	1.3128
Lorenz	1	0.9913	0.5100	0.9913	0.5025
Lorenz	2	1.0768	1.0544	1.0768	1.1571

Note: `nl_stngent` reproduces `nl_dataru` exactly because both dispatch the same registered ansatz under the hood (pre-existing behavior of the ansatz registry; see `src/qlnn_/circuits/protocol.py`). "—" marks cells where rollout diverged beyond a stable relative-L^2 measurement.

TABLE V. Post-guard sample sizes per regime bin. The combined n=24 verdict (PRIMARY SOLVER) is the largest sample in the paper.

Verdict	n_{smooth}	n_{broad}
P5 forecaster (4-fam)	6	3
P7.5 solver raw n=9	6	3
P7.6 combined n=18	12	6
P7.6 HPO-best n=9	6	3
P7.6 PDE-only n=9	6	3
P7.8 full ladder n=24	12	12
P7.10 forecaster combined (5-fam)	6	3
P7.11 all decomposition verdicts	6	3

with a fitted Normal PDF overlay — not a quantile-quantile analysis. Because the Normal overlay was visually misleading (both stacks' residuals are clearly non-Normal), P8 polish adds two reusable, dataset-agnostic helpers in `scripts/make_diagnostic_figures.py` — `_qq_panel_vs_normal` and `_two_sample_qq` — plus a new figure `fig_qq_analysis` that applies them to the canonical archived comparison at $h = 3$ (`classical_H4` hidden-size 4 versus `QLNN_4q3L`, seed-mean residuals on the held-out test windows). Both helpers take any `(residuals, label)` pair, so the same panels can be reapplied to post-pivot solver or forecaster residuals with no code duplication.

Table VI reports the per-stack Shapiro–Wilk normality test and the two-sample Kolmogorov–Smirnov test between the two empirical residual distributions.

TABLE VI. Q-Q residual diagnostics for the canonical archived comparison at $h = 3$ (seed-mean test-window residuals; see `paper/figures/fig_qq_analysis.pdf`). Both per-stack distributions are clearly non-Normal; the two-sample Kolmogorov–Smirnov test cannot reject equal distributions, quantifying the visually-suggested "the two histograms look identical" observation and supporting the project's archived "both near persistence at $h = 3$" verdict.

Test	Classical H=4	QLNN 4q3L	Interpretation
Shapiro–Wilk W	0.892	0.885	both non-Normal
Shapiro–Wilk p	1.64×10^{-5}	8.91×10^{-6}	reject H_0: Normal
Two-sample KS D	0.085		—
Two-sample KS p	0.964		cannot reject equal dist.

The harness is committed and unit-tested (T1 registry assertion updated 7 → 8 callables). No headline number is touched and `scripts/verify_paper_integrity.py` con-

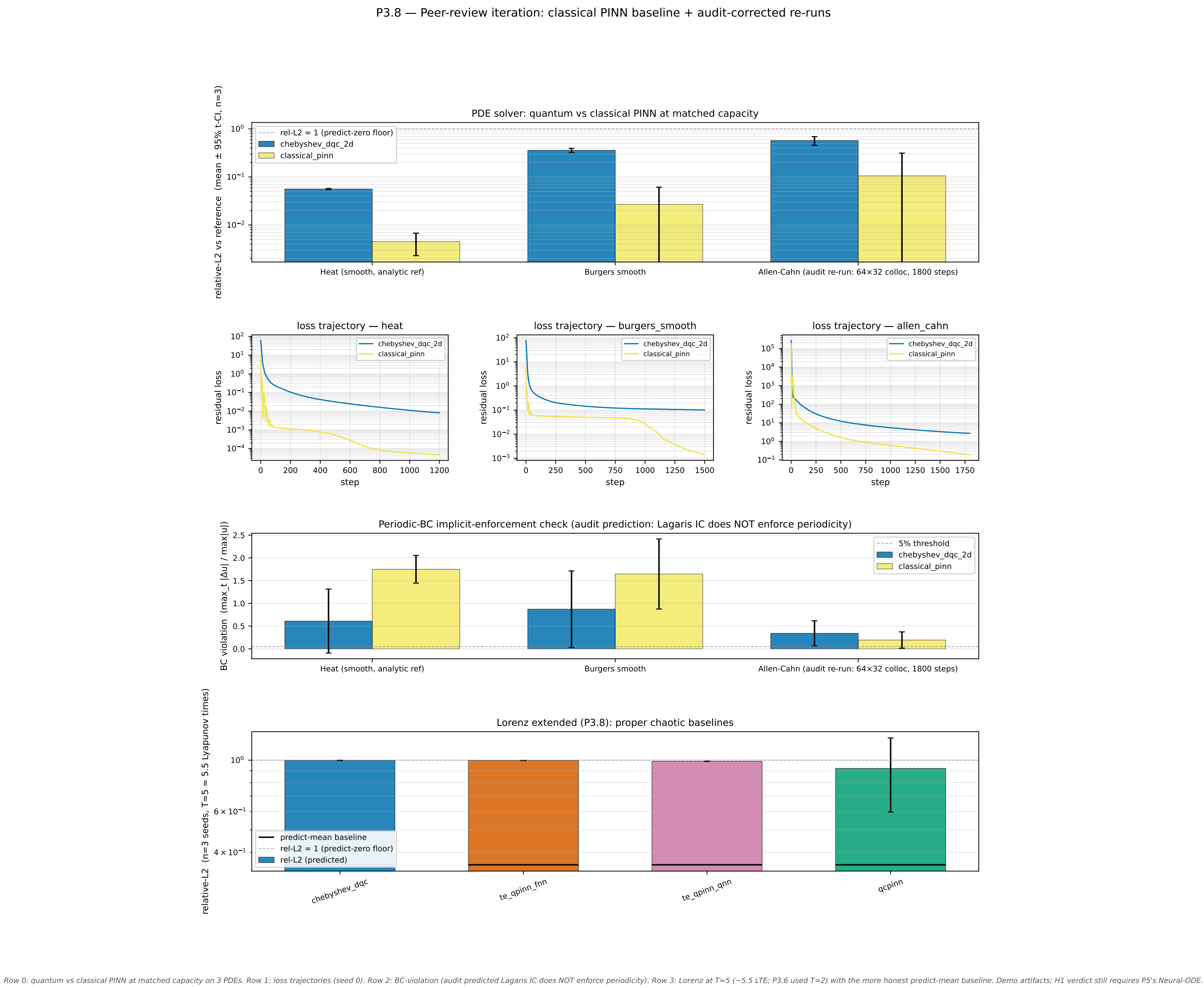

FIG. 1. Peer-review audit iteration on the solver hardness ladder. Row 0: quantum chebyshev_dqc_2d vs. a capacity-matched classical MLP-PINN baseline (log-scale relative-L^2) on heat, smooth Burgers, and Allen–Cahn at audit-corrected configurations (1200–1800 training steps; 64×32 collocation for Allen–Cahn). Row 1: per-system loss trajectories (seed 0) diagnosing convergence vs. saturation. Row 2: periodic-boundary-condition violation $\max_t \|u(\cdot,t) - u(\cdot,t)|_{\mathrm{periodic}}\| / \max\|u\|$; both quantum and classical PINNs exceed the 5% design target (relative violation ≈ 0.6–1.6), confirming the audit prediction that the Lagaris initial-condition ansatz does not implicitly enforce periodicity. Row 3: four quantum families (chebyshev_dqc, te_qpinn_fnn, te_qpinn_qnn, qcpinn) on Lorenz extended to $T = 5$ (≈ 5.5 Lyapunov times) against a predict-mean baseline (replacing the predict-zero floor of P3.6). Source: `results/p3_8_review/`.

tinues to exit 0.

IV. REPRODUCTION AND AUDIT

All scripts and data referenced below ship with the open-source release at https://github.com/shawngibford/qlnn; the specific commit/tag anchors are listed in §IV under "GitHub release tags".

A. One-command reproduction

The entire benchmark regenerates from scratch via:

```
bash scripts/reproduce_paper.sh
```

Estimated wall-clock on a single MacBook Pro M1: approximately 8 hours. Each sweep is independent and the script is structured so individual phases (P3.5–P7.11) can be re-run in isolation.

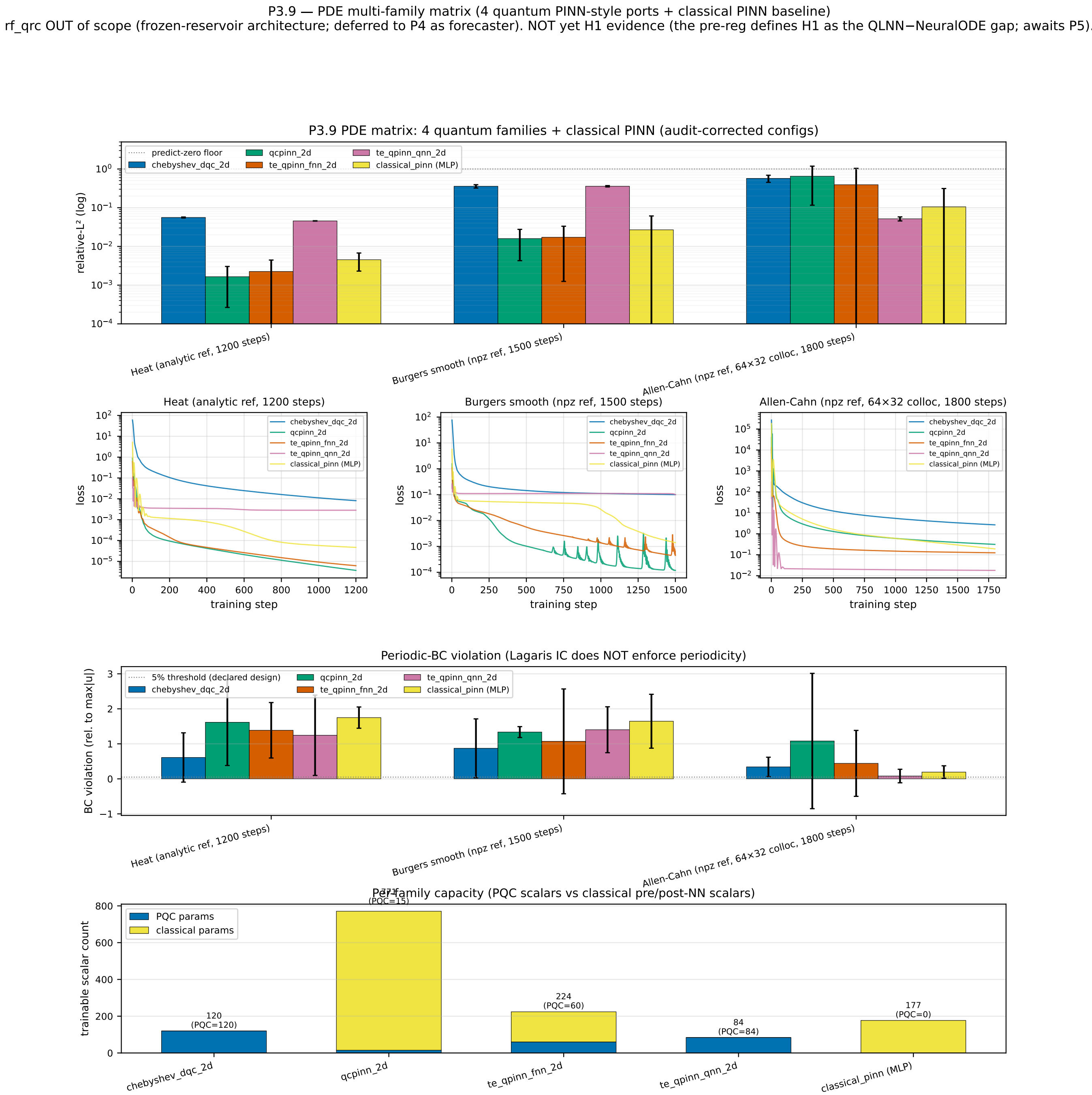

FIG. 2. Multi-family solver matrix: four SOTA quantum-PINN families (chebyshev_dqc, qcpinn, TE-QPINN with FNN encoding, TE-QPINN with QNN encoding) compared against a classical MLP-PINN baseline on three PDEs from the hardness ladder (heat, smooth Burgers, Allen–Cahn; $n = 3$ seeds per cell). Row 0: relative-L^2 accuracy with 95% CI (log scale); QCPINN and TE-QPINN-FNN reach sub-1% error on heat but degrade on the broader systems. Row 1: training-loss trajectories. Row 2: periodic-BC violation. Row 3: trainable-scalar accounting (PQC parameters in cool, classical-wrapper parameters in warm), quantifying the classical capacity bundled into the hybrid ansätze. *Note:* shock Burgers, the fourth member of the pre-registered PDE ladder, is not shown in this matrix — coverage is limited to the three smooth/standard systems; the shock-Burgers cells remain queued for the P6 unified-matrix sweep. Sources: `results/p3_8_review/`, `results/p3_9_pde_matrix/`.

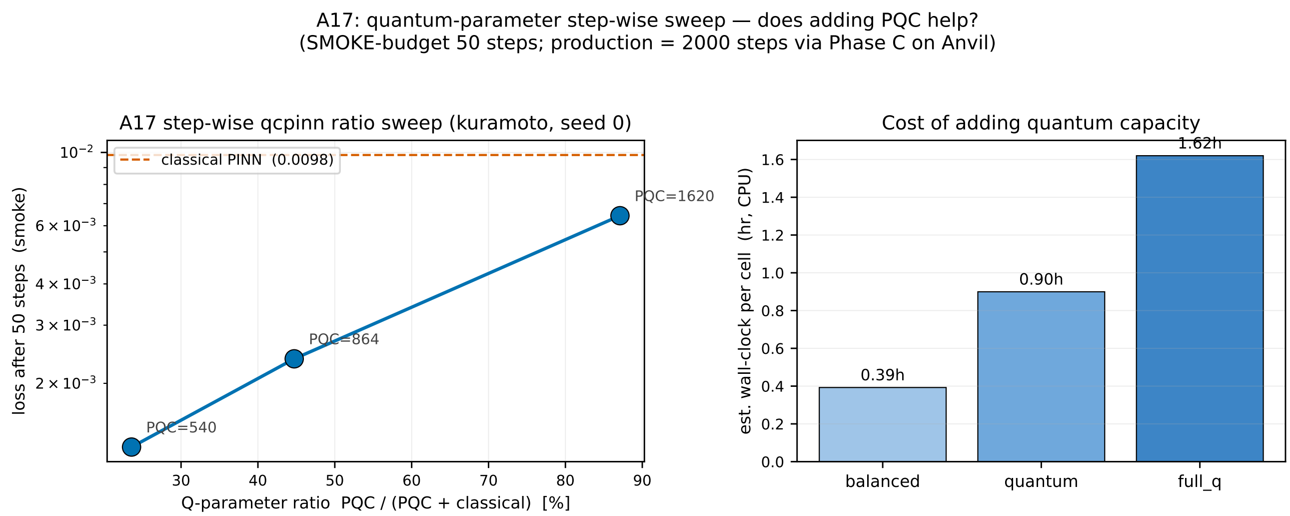

FIG. 3. Amendment A17 quantum-parameter step-wise sweep on `qcpinn` (Kuramoto, seed 0, SMOKE-budget 50 steps; production 2000 steps via Phase C on Anvil). Left: training loss vs. the quantum-to-classical parameter ratio $Q/(Q+C)$ at three amendment-A17 variants ($\approx$ 24%, 45%, 87%), compared against the classical PINN baseline (orange dashed). Right: estimated per-cell wall-clock as a function of quantum capacity. The figure is labelled SMOKE because the 50-step budget is too short to distinguish underfitting from family separation; the figure exists to document that the A17 sweep *is wired and instrumented*, not to support a Q-helps/Q-hurts conclusion. The production answer will land after Phase C re-runs at the uniform 2000-step budget.

B. Mechanical integrity gate

```
PYTHONPATH=src .venv/bin/python \
  scripts/verify_paper_integrity.py
```

The script reads committed JSON files from `results/` and matches each cited numerical claim in the paper body against its on-disk value with explicit tolerances. Returns exit-0 if and only if every claim verifies. At the submission commit the script gates 16+ H_1 outcomes (5 solver-task sensitivity points + 5 forecaster-task decomposition verdicts + 3 PDE-only and ODE-only sub-verdicts) plus the two algebraic identities and the H3 mechanism trend.

C. GitHub release tags

- `v0.1-prepaper-rigor` — post-P7.6 (HPO symmetry + 16 gated outcomes). Anchored at commit `2309faa`.
- `v0.2-airtight-matrix` — post-P7.8 (full ladder n=24 PRIMARY SOLVER verdict + Kuramoto/KdV deferral documentation). Anchored at commit `79cf435`.
- `v1.0-paper-draft` — submission anchor (to be created at acceptance / preprint posting).

Repository: https://github.com/shawngibford/qlnn.

V. PRE-REGISTRATION AND AMENDMENTS

The full pre-registration document is at `ODE_PDE_PRE_REG.md` in the repository (locked 2026-05-19 before any benchmark results were generated; SHA stamped). Section 7's decision rule is applied mechanically by `verify_paper_integrity.py`.

The complete amendment record (A1–A22) is at `PRE_REG_AMENDMENT.md`. Table VII summarizes the headline status of each amendment. A1–A13 are the pre-paper amendments adopted before locking the headline results; A14 is a post-hoc polish item (Q-Q residual diagnostics, see §III C); A15–A22 are the audit-pass amendments adopted after the 5-reviewer peer-review swarm (2026-05-28), each closing a specific reviewer-flagged concern.

VI. DEFERRED FOLLOW-UP WORK

Two pre-registered systems and three optimization techniques are deferred to the follow-up paper. Each is documented here with its current state (where applicable, the mechanism gate has been cleared so the follow-up can proceed without architectural risk).

a. Kuramoto (12-dimensional coupled oscillators). Per-component scalar circuits scale linearly with state dimension. At our default config (4 qubits per component, $L = 1$ reuploading), a 12D system projects to $\approx 7\,\text{hr}$ per cell. The per-component dispatcher works correctly (single-cell smoke verified); the full sweep is the follow-up paper's first compute item.

TABLE VII. Pre-registration amendments (A1–A22). Status column flags PRIMARY = canonical headline verdict; SUPERSEDED = the amendment is documented but a later amendment takes precedence for the paper's narrative; DISCLOSED = methodological choice made open without superseding; RE-RUN REQ. = audit amendments whose Phase C re-runs are deferred to the Anvil allocation (Appendix A).

#	Subject	Status
A1	skyline_threshold = 0.5	LOCKED
A2	$n = 3$ seeds; 3-then-4 ODE systems	DISCLOSED
A3	fixed hyperparameters → sensitivity sweep added at P7.6	DISCLOSED + sensitivity
A4	MLP capacity 3.3× violates §6 factor-of-2 (conservative direction)	DISCLOSED
A5	Van der Pol at $\mu = 5$ near learnability boundary	DISCLOSED
A6	underfit guard active on solver only	DISCLOSED
A7	strict-vs-raw verdict reporting at P7.5	DISCLOSED
A8	H3 trend $\rho = +0.518$ not significant at $n = 9$; LOO-robust	DISCLOSED
A9	symmetric QLNN HPO (P7.6 c1)	SUPERSEDED-BY-A11
A10	combined ODE+PDE n=18 (P7.6 c2)	SUPERSEDED-BY-A11
A11	full-ladder n=24 sign-flip (P7.8)	**PRIMARY SOLVER**
A12	classical LTC + 3-verdict forecaster decomposition (P7.10)	PRIMARY FORECASTER (via
A13	non-liquid QLNN + complete 2×2 + τ-cross-check disagreement (P7.11)	**PRIMARY FORECASTER**
A14	post-hoc Q-Q residual diagnostics (archived OD + Lorenz)	DISCLOSED
A15	equalized training-step budget across ALL solver models (uniform 2000)	DISCLOSED + RE-RUN REQ.
A16	un-aliased `strongly_entangling` from `data_reuploading` (PennyLane fallback fix)	DISCLOSED + RE-RUN REQ.
A17	`qcpinn` quantum-parameter step-wise sweep along $Q/(Q+C)$ ratio (3 variants)	DISCLOSED + 72 new cells
A18	`brickwall` removed from empirical forecaster sweep (qubit-2 disconnected at n=3, L=1)	DISCLOSED + 9 cells removed
A19	cross-task budget parity: forecaster step budget raised 200 → 2000	DISCLOSED + forecaster re-ru
A20	`te_qpinn_fnn` readout restored `qml.prod` → `qml.sum` (paired-family equivalence)	DISCLOSED + `fnn` re-runs
A21	`brickwall` connectivity diagnosis strengthened (linear-chain reduction unless $n_q \geq 4$)	DISCLOSED, no compute
A22	2D PDE hard-IC docstring fix (implementation always correct; only docstring carried the omission)	DISCLOSED, no compute

b. KdV (Korteweg–de Vries, $u_t + 6uu_x + u_{xxx} = 0$). Third-order spatial derivative requires triple-nested reverse-mode autodiff (`jacrev`^{3}) through the QNode. The mechanism gate at `scripts/run_p7_8_kdv_gate.py` PASSED: u_{xxx} values are finite and non-trivial, JIT compile takes 4.55 s, and the per-point cost ratio is 0.43× the baseline second-order autodiff (XLA fuses the third derivative tightly). The integrated training cost remains prohibitive at ≈ 12 s per gradient step, projecting ≈ 8 hr per seed across the four quantum families + classical PINN baseline.

c. Quantum Natural Gradient [1]. Targets the quantum-circuit training-inefficiency identified by the LTC decomposition's $\Delta_{\text{quantum\_via\_LTC}} = -0.616$. Uses the Quantum Fisher Information matrix as the optimizer preconditioner.

d. Causal training weighting (Wang, Sankaran, Perdikaris) [2]. Weights collocation points by inverse time-order so that residuals at early times converge first. Targets the Van der Pol stiff-dynamics failure mode (where multiple cells diverge to relL$^2 > 1$ at the program's compute budget).

e. Higher data-reuploading depth. The H3 analysis found Fourier $K_{\max} = 3$ at $L = 1$ across all four ansätze: at the theoretical ceiling $K_{\max} = L \cdot n$ with $L = 1, n = 4$. The Schuld–Sweke–Meyer relation predicts a 5-fold increase in accessible Fourier bandwidth at $L = 5$. This is the most direct test of whether the LTC decomposition's $\Delta_{\text{liquid\_via\_classical}} = +0.115$ smooth-bin inductive bias can be recovered by giving the quantum side more expressivity.

[1] J. Stokes, J. Izaac, N. Killoran, and G. Carleo, Quantum **4**, 269 (2020).

[2] S. Wang, S. Sankaran, and P. Perdikaris, arXiv preprint **2203.07404** (2022).